Rayan Halder

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.24
ST. Claret School	Class XII (ISC)	2023	93.00%
ST. Claret School	Class X (ICSE)	2021	89.40%

WORK EXPERIENCE

GIAN (Developer)

- Assumed ownership of the production **GIAN** platform as its **sole maintainer**, supporting **302+ higher education institutions** and **5,000+ unique users**, while delivering new features and ensuring platform reliability.
- Designed and implemented full-stack enhancements using **React**, **Tailwind CSS**, **Go**, and **Gin**, improving platform functionality and maintainability, and managing production deployments, bug fixes, infrastructure improvements, and ongoing maintenance for over **one year**, ensuring continuous platform availability and timely feature delivery.
- Automated staging and production deployments by building a **CI/CD pipeline** with **Jenkins**, **GitHub Webhooks**, and **Infisical**, replacing a manual, error-prone deployment process and improving release reliability and led the migration of the platform's file storage from a **local filesystem to Amazon S3**, enabling scalable, durable cloud-based storage and simplifying file management for production workloads.
- Implemented centralized application logging with **Fluentd**, improving observability and accelerating production issue diagnosis while laying the groundwork for future **OpenSearch** integration.

PROJECTS

Zig Standard Library (PR Link)

- Submitted an open-source pull request to the **Zig Standard Library** implementing a **multi-architecture floating-point environment (fenv)**, supporting **14+ CPU architectures**, including x86, x86_64, ARM, AArch64, RISC-V, PowerPC, MIPS, LoongArch64, and s390x.
- Ported and adapted from **musl libc's** fenv implementation to ensure standards-compliant floating-point behavior and broad cross-platform compatibility.
- Worked with low-level systems programming concepts including architecture-specific implementations, compiler standard library development, and IEEE floating-point environment management.
- Received technical reviews and design feedback from **Zig maintainers**, iterating on the implementation through the project's open-source review process and collaborating with project maintainers on a proposed language enhancement to improve the usability and ergonomics of the floating-point environment API.

System Metric Collector (Github Link)

- Developed a high-performance Linux telemetry framework in **C++23** for collecting system-wide CPU performance metrics using **perf_event_open** and **sysfs** interfaces.
- Engineered an asynchronous telemetry backend using **Asio** with **io_uring**, enabling configurable **100 ms** metric collection and low-latency data streaming.
- Designed a modular metrics architecture, allowing new performance counters and telemetry collectors to be added with minimal code changes.
- Optimized the application to operate at **<1% CPU utilization** with approximately **2 MB memory usage** during continuous metric collection and terminal output.
- Implemented efficient binary serialization with **Protocol Buffers** for compact telemetry transmission over multicast UDP, with a messaging abstraction designed for future **ZeroMQ** integration.
- Evaluated **eBPF** as a collection mechanism and selected **perf_event_open** and **sysfs** for broader compatibility and implementation simplicity.

Ray Tracer (Github Link)

- Implemented a **CPU-based ray tracing renderer from scratch in C++**, generating rendered scenes as image outputs with configurable camera positioning.
- Developed a modular rendering pipeline supporting **sphere geometry, ray-object intersection, surface normal calculations, lighting-based shading, and hard shadow generation**.
- Implemented custom vector and geometry operations without external math libraries to handle camera rays, intersections, and lighting calculations.

- Added support for configurable **multiple light sources** with intensity-based illumination calculations to simulate realistic scene lighting.
- Structured the project using **CMake** with an object-oriented architecture for maintainable graphics components.

SKILLS

Languages: C, C++, CMake, SQL (Postgres), Javascript, Typescript, Go, Zig, Rust, Dart

Frameworks & Libraries: React, Svelte, Node.js, Sveltekit, NextJS, react-router, Go-Gin, ASIO, Flutter

Dev Ops: Jenkins, Fluentd, Docker, Opensearch, Infisical, S3

RELEVANT COURSES

Mathematics: Probability Theory, Linear Algebra, Convex Optimization, Applied Statistics

Computer Science: Compilers I, DBMS I, DSA, Theory of Computation, Algorithms